



Faculty of Computing

University of Sri Jayewardenepura

Undergraduate Final Year Project -Proposal Form

Please complete all of the following sections.

Candidate Details

Student Full Name:	Eranga Kavisanka Ariyaratna
Registration No.	FC110589
Examination No.	FC212016

Supervisor Details

Supervisor (1) Name	Mr. Pubudu Jayasekara	Department	Department of Knowledge Engineering and Communication
Supervisor (2) Name	Mr. E. K. D. I. G. Edirimanne	Department	Department of Information Systems Engineering and Informatics

Project Details

Title	A Hybrid RF-Based Off-Grid Communication Network with Opportunistic Cloud Synchronization for Real-Time Traveler Safety and Emergency Response
Objectives	1. To design and implement an RF-based mesh communication network capable of enabling reliable device-to-device communication within a minimum range suitable for off-grid environments. 2. To develop a Flutter-based mobile application integrated with an ESP32-based IoT communication device via USB-OTG for transmitting and receiving text-based messages through the mesh network. 3. To implement and evaluate a real-time SOS emergency alert mechanism capable of propagating emergency notifications across multiple communication nodes with minimal communication delay. 4. To develop an opportunistic cloud synchronization framework utilizing AWS IoT Core and MQTT protocol for transmitting critical communication and emergency data whenever internet connectivity becomes available. 5. To design and evaluate a real-time monitoring dashboard using MySQL and Grafana for visualizing traveler locations, communication activities, and emergency alerts generated within the proposed system.

Second Section (Only accessed by the Supervisors)

Filled by Supervisor	
Decision	• Approved without changes • Approved with minor or major revisions • Rejected, requiring a new submission
Comment	



A Hybrid RF-Based Off-Grid Communication Network with Opportunistic Cloud Synchronization for Real-Time Traveler Safety and Emergency Response

**Faculty of Computing
University of Sri Jayewardenepura**

Final Year Project Proposal

A. G. Eranga Kavisanka Ariyaratna

Registration No. – FC110589

Examination No. – FC212016

2026

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1. Introduction

Effective communication serves an important function in enhancing security and coordinating travelers or hikers especially in areas where regular telecom networks are not available or reliable. Areas such as forests, mountains, wildlife sanctuaries, and even rural areas often lack good network coverage hence traditional phone communication systems may be inefficient during emergencies. Without proper means of communicating to each other or seeking help, individuals may find themselves in danger and unable to perform emergency rescue activities effectively.

Even though a number of existing systems support offline communication capabilities, many lack the ability to integrate with cloud-based systems for monitoring, synchronization of data, and real-time emergency management. Current solutions are also known to offer very little capability for emergency alerting and monitoring the safety of travelers in off-grid and dynamic environments.

The present research aims at developing an RF-based off-grid communication solution that combines with opportunistic cloud synchronization to enhance the real-time safety and emergency management of travelers. The proposed system is comprised of a mobile app developed using Flutter which communicates with the IoT communication device by means of USB-OTG connection. The system employs RF Mesh networking technique for offline communication. Once internet becomes accessible, data will be synchronized to the cloud through AWS IoT Core. The system will also include the capability for SOS emergency alerts along with the real-time monitoring dashboard based on MySQL and Grafana.

This proposed solution is designed to provide an efficient communication model that is reliable, cost-effective, and scalable, in addition to providing contributions to the disciplines of IoT, wireless communications, cloud computing, and emergency response systems. The current study is a continuation of the review-based studies on offline communication systems based on RF technology by applying a hybrid communication system.

2. Problem Statement

The use of cellular network signals may fail to provide reliable communication facilities when travelers and hikers are engaged in outdoor activities in isolated and off-grid settings. This situation may make traditional communication and internet services ineffective as they would be unable to facilitate communication among nearby users. The lack of effective communication means may increase the safety threats and reduce the effectiveness of any emergency measures.

While some offline communication technologies have been developed that facilitate local user-to-user communication using radio frequencies, most of the solutions lack the incorporation of cloud-based applications to support communication needs during emergencies. Moreover, none of the existing applications have developed an efficient means of communicating emergency notifications or sharing information once communication infrastructure is restored.

Moreover, many currently available methods are plagued with challenges such as limited communication range, inability to scale well, high costs of implementation, and poor integration of IoT and cloud-based models. There is, therefore, an urgent need for a scalable and affordable hybrid communication system that can offer reliable communication services in off-grid situations while at the same time being able to facilitate opportunistic cloud synchronization and emergency management capabilities. The objective of this research is to develop a hybrid communication system based on radio frequency technology that will offer off-grid communication services along with opportunistic cloud synchronization.

3. Aim / Objectives

Aim

To design and develop a hybrid RF-based off-grid communication network integrated with opportunistic cloud synchronization for improving traveller safety and emergency response in environments with limited or no cellular connectivity.

Objectives

1. To design and implement an RF-based mesh communication network capable of supporting reliable device-to-device communication in off-grid environments.
2. To develop a mobile application integrated with an IoT communication device via USB-OTG for transmitting and receiving messages through the mesh network.
3. To implement a real-time SOS emergency alert mechanism capable of propagating emergency notifications across multiple communication nodes.
4. To develop an opportunistic cloud synchronization framework utilizing AWS IoT Core to transmit critical information whenever internet connectivity becomes available.
5. To design and evaluate a monitoring dashboard using MySQL and Grafana for visualizing traveller status, communication activities, and emergency alerts in real time.

4. Method/s

The research would use the Design Science Research Methodology (DSRM) as the main research methodology for this research. This research would involve designing, development, and evaluation of the design for a hybrid communication system for traveller safety and emergencies in disconnected environments.

This system will comprise three major layers which include the Edge Layer, the Communication Layer and the Cloud layer

Edge layer comprises the Flutter-based mobile application that would be linked to the ESP32-based IoT communication device via USB-OTG connectivity. The IoT device would employ LoRa-based RF communication modules for enabling low-power and long-distance communication among nearby users within disconnected environments without cellular networks.

The Communication Layer will incorporate mesh networking protocols that rely on the RF technology to facilitate multi-hop communication between several devices. The Communication Layer will ensure messages and emergency alerts pass through adjacent nodes to expand coverage in situations where there is no access to grid-based communication networks. Moreover, the SOS emergency alert system will be incorporated to enable quick sharing of important messages.

The Cloud Layer will make use of the internet whenever it becomes accessible to achieve synchronization of cloud resources in an opportunistic manner. AWS IoT Core will provide MQTT communication between gateway devices and cloud servers. Incoming messages will then be processed using AWS Lambda and stored in a MySQL database. Finally, Grafana will provide visualization of traveler-related information and emergency alerts.

The development stage of the project entails the development of the mobile application, configuration of IoT communication devices, implementation of mesh communication algorithms, use of cloud services, and development of monitoring dashboards. Performance evaluation of the proposed system will take place by

analysing metrics like delivery ratio, communication latency, synchronization delay, reliability of emergency alerts propagation, and system effectiveness under disconnected networks.

5. Research Contribution

The proposed research will contribute to both theoretical knowledge and real-world applications in the areas of Internet of Things (IoT), wireless communications, cloud computing, and emergency response mechanisms.

Firstly, the research proposes a new communication framework by employing the combination of RF mesh network and opportunistic cloud communication. In contrast to current communication frameworks that use internet connection only, the proposed framework facilitates communication under disconnected networks by synchronizing important data to cloud-based services in case there is availability of connections.

Secondly, the research provides an innovative approach towards solving the problems faced during emergencies caused due to disconnected internet services. In addition, this research will also provide an innovative way of providing off-grid communication facilities for travellers and hikers.

Thirdly, the design makes use of a real-time SOS emergency warning system that is able to distribute emergency warnings through different communication points within the mesh network. In addition to improving emergency response services, this will ensure improved safety of users within remote locations.

Fourthly, the inclusion of AWS IoT Core, MySQL, and Grafana allows for the implementation of centralized monitoring and visualization features into the proposed design. With the help of these technologies, it will be possible to monitor in real time the activity of travellers as well as communication events and emergency situations.

Finally, this project offers empirical evidence on the reliability, effectiveness, scalability, and feasibility of RF-based hybrid communication solutions within disconnections. The results of this research may be helpful for future investigations concerning the design of disaster response systems, rural communications, and IoT safety solutions.

6. Project Timeline

No.	Task	Duration (Weeks)	Start Date	End Date
1.	Literature refinement and proposal preparation	2 weeks	26 May 2026	08 June 2026
2.	Hardware procurement and RF communication setup	2 weeks	09 June 2026	22 June 2026
3.	Mobile application development and IoT integration	3 weeks	23 June 2026	13 July 2026
4.	Mesh networking and SOS alert implementation	3 weeks	14 July 2026	03 August 2026
5.	AWS IoT Core integration and cloud synchronization	2 weeks	04 August 2026	17 August 2026
6.	MySQL database and Grafana dashboard development	2 weeks	18 August 2026	31 August 2026

7.	System testing and performance evaluation	3 weeks	01 September 2026	21 September 2026
8.	Thesis writing and documentation	Continuous	26 May 2026	February 2027

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I hereby certify to the best of my knowledge that the details mentioned above are true and accurate.

Signature of Candidate

Date



28 May 2026

I/ We hereby certify to the best of my/ our knowledge that the details mentioned above are true and accurate.

Signature of Supervisor (s)

Date



2026/05/29



Pubudu S Jayasekara

29 / 05 / 2026